

Random-Walk and Efficiency Tests of Central European Equity Markets

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Abstract

The existence of weak-form efficiency in the equity markets of the three main Central European transition economies (the Czech Republic, Hungary, and Poland) is examined for the period July 1995 through September 2000, using weekly Investable and Comprehensive indexes developed by the International Finance Corporation. Several different approaches are used. Univariate and multivariate tests provide some evidence that stock prices in these exchanges exhibit a random walk, which would support weak-form efficiency. This differs in some cases from studies using data for the initial years of these markets. The variance ratio test of Lo and MacKinlay (1988) yields somewhat mixed results concerning the random walk properties of the indexes. A model-comparison test compares forecasts from a NAÏVE model with ARIMA and GARCH alternatives. Results from the model-comparison approach are consistent in rejecting the random-walk hypothesis for the three Central European equity markets.

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1. Introduction

Following the fall of Communism in Central Europe, a number of stock exchanges that had been closed in the postwar era reopened. The first to do so was the Budapest stock exchange (BSE) on June 21, 1990, followed by the Warsaw stock exchange (WSE) on April 16, 1991. On April 6, 1993, the Prague stock exchange (PSE) started its trading. Other equity markets have also opened as part of the transformation of the region's countries into free-market economies. Concerns had been expressed from the outset of transition that equity markets would have difficulties functioning due to such problems as low liquidity and high volatility (e.g., Lipton and Sachs, 1990; Tirole, 1991). At the present time some of these exchanges have been functioning for nearly a decade, and a number of transition countries are presently endeavoring to bring their economies, markets, and institutions into line with Western European models in the hope of joining the European Union (EU) and the European Monetary Union (EMU). The Czech Republic, Hungary, and Poland, currently associate members of the EU, are the three countries in Central Europe most likely to become full members and to join the EMU sometime in the first decade of this century.

An important empirical question in this context is whether the stock markets of the Czech Republic, Hungary, and Poland are efficient. The main purpose of this paper, therefore, is to investigate the behavior of these three exchanges in light of the Efficient Market Hypothesis (EMH). In particular, we explore the issue of whether these markets have exhibited a movement toward efficiency once they became established, acquired a degree of experience, and introduced regulatory and institutional frameworks similar to those of more developed countries. In

addition, we briefly inquire whether barriers to foreign investment in these markets may have had an impact on their efficiency.

To investigate whether stock prices on these exchanges are efficient we use several index series and a variety of tests. Univariate methods, including the Augmented Dickey-Fuller (ADF) test (1979) and the variance ratio test of Lo and MacKinlay (1988), as well as multivariate tests, including the Johansen (1988) approach and the Granger causality test (1969), are applied for the period July 5, 1995, through September 27, 2000. Further, a model-comparison approach, which involves comparing the forecasting results produced by a NAÏVE (random walk) model with those of autoregressive integrated moving average (ARIMA) and generalized autoregressive conditional heteroskedastic (GARCH) models, is also performed. Although the serial correlation-based tests largely support a conclusion that these markets are weak-form efficient, the results of comparing forecasts of alternative models are consistent in rejecting the random-walk hypothesis.

The paper is organized as follows. The main characteristics of the Central European stock markets are described in the next section. Section 3 provides a review of the relevant literature, and the data are described in Section 4. The methodologies utilized in this paper are explained in Section 5, and Section 6 presents the results. We conclude in Section 7.

2. Market Characteristics

Basic information about the three Central European stock markets is presented in Table 1. The Warsaw exchange has been the largest in terms of market capitalization in the recent past, followed by the BSE and the PSE. The BSE initially lagged in terms of turnover ratios and trading values but has overtaken the others.

The exchanges reflect the different privatization strategies pursued by the three countries. Trading on the BSE began two years before the initiation of a voucher program in Hungary. Trading of vouchers was permitted under Hungarian privatization law, bringing with it a widespread participation and experience for traders in these instruments.¹ The BSE has also been characterized by a substantial presence of foreign ownership through direct institutional investment and cross-listing of shares on exchanges in Vienna, Stuttgart, Munich, and elsewhere. The WSE opened trading in the shares of only five firms, which had been privatized at the end of 1990. The number of listed firms has grown slowly, as a result of a gradual approach to privatization. Growth in the number of firms listed on the BSE has been similarly gradual. In contrast, the Prague exchange opened contemporaneously with the first of several mass waves of privatization, which led to a large number of firms being listed through 1996. Since then, the majority of those firms have been delisted. In each of these countries institutions such as securities commissions have been introduced, and there has been an evolution of laws and regulations, including requirements for disclosure of information relevant to share prices.

3. Literature Review

Some studies have produced evidence against the random walk hypothesis, showing that stock returns contain predictable elements. Much of this work has centered on the world's largest stock markets, including the United States, developed economies in Europe, and Japan (see, among others, Poterba and Summers, 1988; Lo and MacKinlay, 1988; and Fama and French, 1988). Subsequent research by Campbell *et al.* (1997) presents updated results of the Lo and MacKinlay (1988) study, showing no rejection of the random-walk hypothesis using a value-weighted portfolio of weekly returns.

More recently, mixed evidence on the random-walk hypothesis has been found for emerging markets in Latin America (Urrutia, 1995; Grieb and Reyes, 1999) and in Asia (Ayadi and Pyun, 1994; Huang, 1995; Chang and Ting, 2000). There are several studies covering the startup years of some Central European stock markets. Using daily BUX data for 1992-1997 produced by the BSE, Chun (2000) finds that the Hungarian market is weakly efficient. The tests he employs include the Augmented Dickey-Fuller (ADF) (1979) test and the Lo and MacKinlay (1988) variance ratio (VR) method. He also provides some evidence, using the PX and the WIG indexes, that the Czech and Polish stock prices do not follow a random walk during that time period. Nivet (1997) examines the performance of the Warsaw exchange using daily and weekly index data from the WIG for the period 1991-94. On the basis of autocorrelation coefficients, he concludes that the model of a random walk for the Warsaw stock market cannot be supported for those years.

The present research enhances the literature by providing a contrast to the prior results. We examine the behavior of the Czech, Hungarian, and Polish stock markets using weekly index data from the International Financial Corporation (IFC) Emerging Markets Data Base and focus on the more recent performance of these markets, from 1995 to 2000. By eliminating the early years when the Central European exchanges were just being established, we are able to provide a picture of their evolution toward the model of markets in more developed countries. Our serial-correlation-based tests support the findings of Chun (2000) on the Hungarian market, but our model-comparison tests lead to a rather different conclusion. With respect to the Czech and Polish markets, serial-correlation-based tests of their relatively recent performance contrast with the findings of inefficiency in Chun (2000) and Nivet (1997), but our model-comparison results fail to support a random-walk model of stock price behavior.

4. Data

We use total return data for stock indexes for the Prague, the Budapest, and the Warsaw exchanges for the period July 5, 1995, through September 27, 2000, a set of 1376 daily observations. The starting date is determined by data availability for the indexes. These indexes were obtained from the Emerging Markets Database, maintained by the International Finance Corporation (IFC). Individual securities within these markets are classified by the IFC as either restricted, for securities that have some barrier to foreign ownership, or unrestricted, indicating they may be freely purchased by foreign investors. Thus, the IFC provides two market-capitalization-weighted indexes for each country: an Investable index which covers unrestricted securities only and a Comprehensive index which includes both restricted and unrestricted securities. These indexes are constructed to represent a broad indicator of market movements; the target aggregate market capitalization is 60% to 75% of the total capitalization of all listed shares on the local stock exchange. The IFC Investable index is constructed in the same way for a subset of the Comprehensive index constituents that the IFC has determined to be investable, that is, available to foreign institutional investors.² They provide a better means to evaluate the performance of emerging stock markets than do local stock price indexes, which are calculated in varying ways and are based on stocks selected by different criteria. Following Lo and MacKinlay (1988), we conduct our tests using weekly series for each index in order to avoid the problems of biases due to non-trading, bid-ask spreads, and nonsynchronous prices which affect daily sampling. There are 274 weekly observations, based on the closing value for Wednesday of each week.

5. Methodology

5.1 Univariate Tests

A requirement for a random walk is that the time series must contain a unit root. Therefore, we test initially for the presence of a unit root in the log level series. The two methods applied are the Augmented Dickey Fuller (ADF) test, an extension of the Dickey and Fuller (1979) method, and the Phillips-Perron (PP) test. The ADF test uses an ordinary least squares (OLS) regression of the first differences of the series against the series lagged once, as well as lagged difference terms, with optional constant and time-trend terms:

$$\Delta x_t = a_0 + a_1 t + \gamma x_{t-1} + \sum \beta_i \Delta x_{t-i+1} + e_t \quad (1)$$

In this equation Δ is the first difference operator, a_0 is an intercept, $a_1 t$ is a linear time trend, e_t is an error term, and i is the number of lagged first-differenced terms such that e_t is white noise. The test for a unit root has the null hypothesis that $\gamma = 0$. Phillips and Perron (1988) developed a generalization of the Dickey-Fuller test:

$$x_t = b_0 + b_1 x_{t-1} + b_2(t - T/2) + \mu_t \quad (2)$$

A unit root is a necessary but not a sufficient condition for a random walk. Unit root processes can have predictable elements, but a random walk for stock prices means that returns must be uncorrelated. Therefore, we examine the returns for indications of autocorrelation, and we test the random walk hypothesis directly by means of the variance ratio method of Lo and MacKinlay (1988). The basis for this test is the fact that the variance of increments in a random-

walk process is linear in the sampling interval. If prices follow a random walk, the return variance should be proportional to the return horizon. The variance ratio statistic, $VR(k)$, is defined as:

$$VR(k) = \frac{\sigma^2(k)}{\sigma^2(1)} \quad (3)$$

where $\sigma^2(k)$ is the unbiased estimator of $1/k^{\text{th}}$ of the variance of the k -period returns and $\sigma^2(1)$ is the unbiased estimator of the variance of one-period returns. (For fuller details see Lo and MacKinlay, 1988).

Lo and MacKinlay derived asymptotic standard normal test statistics for their variance ratio. The modified test statistics presented here are from Liu and He (1991). The first test statistic, $z(k)$, is developed under the maintained hypothesis of homoskedasticity:

$$z(k) = \frac{VR(k) - 1}{[\phi(k)]^{1/2}} \sim N(0,1) \quad (4)$$

where $\phi(k) = [2(2k - 1)(k - 1)] / k(nk)$. Since variances of most stock returns are conditionally heteroskedastic with respect to time, Lo and MacKinlay (1988) also derive a heteroskedastic version of the test in which the second test statistic, $z^*(k)$, is given by:

$$z^*(k) = \frac{VR(k) - 1}{[\phi^*(k)]^{1/2}} \sim N(0,1) \quad (5)$$

where

$$\phi^*(k) = \sum_{j=1}^{k-1} \frac{2(k-j)}{k} \delta(j)$$

and

$$\delta(j) = \frac{\sum_{t=j+1}^{nk} (X_t - X_{t-1} - \mu)^2 (X_{t-j} - X_{t-j-1} - \mu)^2}{\sum_{t=1}^{nk} [(X_t - X_{t-1} - \mu)^2]}$$

In Monte Carlo simulations Lo and MacKinlay (1989) show that the asymptotic distribution of $z^*(k)$ performs well in finite samples and that the VR test is more reliable than traditional tests such as Dickey-Fuller and Box-Pierce.

5.2 Multivariate Tests

It is also useful to determine whether the three Central European markets are jointly efficient or whether one or more markets could contain information important in forming forecasts of the others, indicating market inefficiency. For this purpose we employ the Johansen procedure (Johansen, 1988; Johansen and Juselius, 1990) to test for the possibility of a cointegrating relationship between the markets.

The Johansen procedure relies on the relationship between the rank of a matrix and its characteristic roots. Let X_t be a vector of n time series variables, each of which is integrated of order (1), and assume that X_t can be modeled by a vector autoregression (VAR):

$$X_t = A_1 X_{t-1} + \dots + A_p X_{t-p} + \varepsilon_t \quad (6)$$

Rewrite the VAR as

$$\Delta X_t = \Pi X_{t-1} + \Sigma \Gamma \Delta X_{t-1} + \varepsilon_t \quad (7)$$

where $\Pi = \Sigma A_i - I$ and $\Gamma_i = -\Sigma A_j$. Under the assumption that each series X_t is nonstationary, the rank of Π will be less than n . In the event that the rank of Π is zero, we can model the system as a standard VAR in first differences.

The tests used to determine the rank of Π are the trace test and the maximum eigenvalue test:

$$\lambda_{\text{trace}}(r) = -T \sum \ln(1 - \lambda_i) \quad (8)$$

and

$$\lambda_{\text{max}}(r, r+1) = -T \ln(1 - \lambda_{T+1}) \quad (9)$$

where λ_i = the estimated values of the characteristic roots (eigenvalues) obtained from the estimated Π matrix, r is the number of cointegrating vectors, and T = the number of usable observations. The trace statistic tests the null hypothesis that the number of distinct cointegrating vectors is less than or equal to r against a general alternative. The λ_{max} statistic uses the null that there are exactly r versus $r+1$ cointegrating vectors.

5.3 Model-comparison tests

Since there is debate over the power of autocorrelation-based tests, it is also useful to employ additional testing methods (Cuthbertson, 1996; Darrat, 2000). The point of a model-

comparison approach is that if stock prices follow a random walk, a random walk (NAIVE) model should not be out-predicted by other models. To implement this approach we estimate ARIMA and GARCH models of the index series, using a base sample which omits the last thirteen weeks, for which the forecasts are formed. A dynamic forecasting approach is used, in which the base sample is enlarged successively by the forecast values produced from the model rather than the actual price values. The one-week-ahead forecasts of log stock prices are then compared using the root-mean-squared error (RMSE) and the mean-absolute-value (MAE) statistics.

While the RMSE and the MAE statistics will indicate which model performs “better,” it will not determine whether one model is significantly better than the alternatives. Therefore, we utilize a method described in Donaldson and Kamstra (1966). In this approach model (b) is superior to another model (a) if b’s forecasts significantly explain a’s forecasting errors and model (b) incorporates relevant information ignored by model (a). Using t-statistics, the significance of the β and γ coefficients is tested in the following two regression equations:

$$(P_{at} - P_t) = \beta P_{bt} + e_t \quad (10)$$

$$(P_{bt} - P_t) = \gamma P_{at} + \varepsilon_t \quad (11)$$

where $(P_{at} - P_t)$ and $(P_{bt} - P_t)$ are the forecasting errors from model (a) and model (b), respectively; P_{at} and P_{bt} are the forecasts of the two models, and e_t and ε_t are random errors. The null hypothesis is that neither model outperforms the other. If $\beta \neq 0$ but $\gamma = 0$, the null hypothesis is rejected in favor of the alternative hypothesis that model (b) outperforms model (a). If $\gamma \neq 0$ but $\beta = 0$, model (a) is superior to model (b). If both β and γ are not significant, or both

are significant, the null hypothesis is not rejected. If any model outperforms the NAÏVE model, we can conclude that stock prices in the Central European markets do not follow a random walk process.

6. Results

6.1 *Univariate Tests*

In Table 2 we show the summary statistics for the Investable and the Comprehensive index returns for the three countries. Evidence of skewness and kurtosis is present, which is typical for stock returns, and the Jarque-Bera statistic indicates that each of the returns series is not normally distributed.

Table 3 presents Ljung-Box Q statistics for the weekly returns. For comparison with previous research we also show the Q statistics on daily returns. The Q statistics test the joint hypothesis that all the calculated autocorrelation coefficients are zero. There are several interesting points to observe. First, the daily return data show high values for the Ljung-Box Q statistics for the entire 1995-2000 period for all three countries, indicative of potential autocorrelation. These results are similar to those reported by Chun (2000) for the Hungarian BUX index over the 1992-1997 period, and with the correlograms calculated by Nivet (1997) for the Polish WIG index for 1991-1994. However, when we break down the series into three approximately equal sub-periods, an evolution in the pattern of Q statistics emerges. The highest Q statistics appear in the early three years (mid 1995 through mid 1998) and become mostly insignificant in the final two years (mid 1998 through mid 2000). These results indicate that over time all three markets are moving in the direction of lower levels of autocorrelations in returns. In comparison, the weekly series Q statistics show little evidence of autocorrelation.

We note a similar pattern of declining autocorrelations over time, although most of the results using weekly data are not significant. The differences in results indicate the possible presence of spurious autocorrelation in daily data. Second, for the daily returns series the Q statistics tend to be more significant for the Comprehensive series than for the Investable series. This pattern is much weaker for the weekly return series. It appears that the Investable indexes, which contain only stocks that are available to foreign as well as domestic investors, show much less evidence of autocorrelation at the daily level, although most of the effect disappears at the weekly level. If the higher levels of autocorrelation at the daily level are indeed spurious, there appears to be little difference between the two index series in terms of autocorrelation.

Given the possible spurious autocorrelation problems of daily data, we pursue further tests only on the weekly indexes. Testing for a unit root is a necessary precondition to proceeding to the variance ratio test. Tables 4 and 5 present results for ADF and PP unit root tests and for the VR statistics on the weekly indexes, respectively. In all cases we find that the Investable and the Comprehensive series are $I(1)$. The VR test results are shown for both the assumption of homoskedasticity and of heteroskedasticity. The index series were tested for heteroskedasticity, which was confirmed in each case. Consequently, the statistics $z^*(k)$ are relevant. They fail to reject the random walk hypothesis for either the Investable or the Comprehensive indexes for any of the three markets. The number of lag periods has been limited to ten for the weekly data since the power of the VR test declines as k becomes large relative to the data set (Lo and MacKinlay, 1989). For all series the differences between $z(k)$ and $z^*(k)$ appear to reflect primarily variance clustering, since correcting for heteroskedasticity consistently reduces the size of the variance ratio statistics. Chun (2000) argues that the significant positive variance ratios he found for daily data from the Czech and the Polish

exchanges were not likely due to positive autocorrelations caused by infrequent trading, which would induce positive spurious correlations. Using a model of nontrading by Lo and MacKinlay (1988), Chun points out that for his indexes to produce the degree of positive autocorrelation implied by his variance ratios, over 40 percent of the securities in the Czech and Polish markets would have to be untraded during a one-week period. If frequency of trading is thus not likely to be an important factor in the differences between our results and those of Chun, then other elements, such as improved regulatory and institutional structure over time, may be a possible explanation.

6.2 Multivariate Tests

Evidence from a univariate analysis thus supports weak-form efficiency in the three exchanges. However, another element to be considered is the possibility that a market may contain information that is useful in creating forecasts of another. Therefore, the Johansen test was applied to these markets. Table 6 shows the results for both the Investable and the Comprehensive indexes. In both cases the conclusion is that the rank of the cointegrating vector is zero. This may be interpreted as indicating that there is no common stochastic trend shared by the three markets and that returns from one are not predictable in terms of information in another.

Given the absence of a cointegrating relationship, we turn to a test of Granger causality, using the return series without an error-correction term. The Granger method (Granger, 1969) seeks to determine how much of a variable, Y , can be explained by past values of Y and whether adding lagged values of another variable, X , can improve the explanation. Y is said to be

“Granger-caused” by X if X helps predict Y, that is, if the coefficients on the lagged X’s are statistically significant, as measured by an F test. The Granger test takes the form:

$$Y_t = \alpha_0 + \sum \alpha_i Y_{t-i} + \sum \beta_j X_{t-j} + \varepsilon_t$$

$$X_t = a_0 + \sum a_i X_{t-i} + \sum b_j Y_{t-j} + \mu_t \quad (12)$$

The results in Table 7 show a lack of causality in either direction between the Czech and the Hungarian markets. However, the Investable index indicates causality running from the Hungarian to the Polish market. Both indexes produce significant evidence for causality from the Czech to the Polish market. This implies that knowledge of the past return behavior in one market is unlikely to improve forecasts of returns in another, except for some evidence of causality running from the Czech and Hungarian to the Polish market. However, this information would not necessarily be economically useful.

6.3 Model-comparison approach

Table 8 reports the RMSE and MAE statistics for one-week ahead forecasts for both the Comprehensive and the Investable indices for each of the three Central European markets. These results indicate that, in general, both the ARIMA and the GARCH models produce smaller forecasting errors than the NAÏVE model in each market. Equations 10 and 11 were then used to test for significant differences in forecasting ability. The results are shown in Table 9. They indicate that the ARIMA and the GARCH models are significantly superior to the NAÏVE model, thus providing evidence against the random-walk hypothesis for these Central European equity markets.

7. Conclusions

This study has applied an array of tests to determine whether recent stock price behavior in the three major CE equity markets is appropriately modeled by a random walk, indicating weak-form efficiency. Univariate, serial-correlation-based tests largely failed to reject the random walk. Our results are consistent with those of Chun (2000) for Hungary, although they differ from his variance ratio results for the Czech Republic and Poland. We also differ with the autocorrelation results of Nivet (1997) for the WSE. The inconsistencies in findings appear to be due to differences in frequency of data, time periods, and indexes. Our weekly data give very little evidence of serial correlation, and both daily and weekly series show a pattern of reduced correlation in very recent years. The multivariate tests used here provide somewhat mixed evidence, with the Johansen cointegration test indicating an absence of any long-term common stochastic trend driving these markets, while there does appear to be some Granger-causality running from the Czech and Hungarian markets to the Polish exchange.

Additionally, given the debate about the power of some of these tests (Poterba and Summers, 1988; Cuthbertson, 1996), we applied a model-comparison approach. Our results rejected the random-walk model in favor of models which made use of dependency in the returns. Overall, it may be concluded that there is strong evidence that stock prices in these markets do not follow a random-walk process but do exhibit some dependency which is captured by ARIMA and GARCH models. It should be noted that a rejection of the random-walk hypothesis does not in itself necessarily negate market efficiency (Summers, 1986). It may only indicate that the results are inconsistent with the particular martingale process of a random walk.

Some possible explanations can be provided for the rejection of the random-walk model for the Central European markets. One obvious possibility is nonsynchronous trading in these

small markets. Infrequent trading could produce some predictability in market returns if new information is not instantly embedded in stock prices and/or a high percentage of stocks remain inactive. Nonsynchronous trading may be a contributing factor, but is unlikely to be a sufficient explanation of our results by itself. A second possible explanation is market imperfections in these transition economies that interfere with the rapid processing of information. These equity markets have been in existence for less than ten years, and although Western-style regulations and practices have been gradually introduced, it is quite plausible that further progress remains to be made.

In investigating market data for evidence of random-walk behavior and market efficiency it is helpful to employ an array of tests. In particular, the use of the model-comparison approach led us to the conclusion that the Hungarian equity market is not necessarily an anomaly among the Central European markets, as implied in Chun (2000). Rather, all three markets show similar patterns of somewhat mixed evidence. Problems of nonsynchronous trading in these small markets, as well as market imperfections, appear to continue to affect the behavior of stock prices in each country.

Notes

¹For a discussion of the efficiency of the Hungarian voucher market see Chun (2000).

²The number of stocks included in the IFC comprehensive indexes are: the Czech Republic, 37, Hungary 15, Poland, 35. For the investable indexes the numbers are: the Czech Republic, 13, Hungary, 13, and Poland, 34. (IFC, *Emerging Stock Markets Factbook*, 1999).

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Table 1: Annual Data for Central European Stock Exchanges, 1995-1998

	1995	1996	1997	1998
Panel A: Czech Republic				
Capitalization (US\$ m)	15,664	18,077	12,786	12,045
Turnover ratio (%)	23.2	50.3	47.9	38.2
Trading value (US\$ m)	3630	8431	7055	4741
Number of firms	1635	1588	276	261
Panel B: Hungary				
Capitalization (US\$ m)	2,399	5,273	14,975	14,028
Turnover ratio (%)	14.8	31.1	51.3	111.3
Trading value (US\$ m)	355	1,641	7,684	16,135
Number of firms	42	45	49	55
Panel C: Poland				
Capitalization (US\$ m)	4,564	8,390	12,135	20,461
Turnover ratio (%)	60.7	66.0	65.7	54.7
Trading value (US\$ m)	2,770	5,538	7,977	8,921
Number of firms	65	83	143	198

Source: International Finance Corporation, *Emerging Stock Markets Factbook 1999*. Turnover ratio is the total value traded divided by annual market capitalization.

Table 2: Summary Statistics of IFC Investable and Comprehensive Index Returns Series (Weekly) for the Czech Republic, Hungary, and Poland

Statistic	Country					
	Czech Republic		Hungary		Poland	
	IFCI	IFCC	IFCI	IFCC	IFCI	IFCC
Mean	0.0010	0.0008	0.0054	0.0060	0.0027	0.0030
Median	0.0025	0.0030	0.0038	0.0040	0.0024	0.0026
Maximum	0.1829	0.0801	0.2059	0.1939	0.1463	0.1393
Minimum	-0.1598	-0.1385	-0.1781	-0.1373	-0.1983	-0.1983
Stan. Dev.	0.0388	0.0297	0.0485	0.0466	0.0447	0.0423
Skewness	-0.1064	-0.6608	-0.0159	-0.0777	-0.1536	-0.0832
Kurtosis	6.0989	4.8811	5.3876	4.8750	4.8428	4.6644
Jarque-Bera	110.9560	60.1176	65.5702	40.2662	39.1859	31.8189
Probability	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Index returns are estimated as the log-relative of weekly prices for July 5, 1995, to September 27, 2000. The Jarque-Bera statistic tests the null hypothesis of a normal distribution and is distributed as a χ^2 with 2 degrees of freedom.

Table 3: Ljung-Box Q Statistics of the IFC Investable and Comprehensive Index Returns Series (Weekly and Daily) for the Czech Republic, Hungary, and Poland

Period	Weekly			Period	Daily		
	Q(5)	Q(10)	Q(20)		Q(5)	Q(10)	Q(20)
Panel A: Czech Republic							
a. IFC Investable Indexes							
1995-2000	7.99	12.38	19.40	1995-2000	24.77**	41.07**	59.90**
1995-1997	8.87	14.35	21.91	1995-1996	5.65	13.49	20.19
1998-2000	6.23	9.33	12.90	1996-1997	18.27**	32.23**	51.71**
				1997-1998	11.32*	15.04	26.51
				1998-1999	13.53**	17.78	25.61
				1999-2000	3.59	10.72	25.72
b. IFC Comprehensive Indexes							
1995-2000	9.94	13.22	27.05	1995-2000	92.74**	107.88**	129.45**
1995-1997	9.15	11.81	34.93*	1995-1996	17.18**	21.48**	31.18
1998-2000	4.13	6.19	27.30	1996-1997	61.70**	78.10**	94.30**
				1997-1998	24.47**	34.41**	44.50**
				1998-1999	24.25**	26.64**	30.82
				1999-2000	4.56	9.43	25.73
Panel B: Hungary							
a. IFC Investable Indexes							
1995-2000	8.36	17.04	20.84	1995-2000	21.90**	32.73**	42.21**
1995-1997	2.83	8.98	14.69	1995-1996	48.33**	52.84**	71.57**
1998-2000	7.93	15.73	19.72	1996-1997	14.58**	20.09**	28.95
				1997-1998	0.78	11.11	24.14
				1998-1999	14.84**	15.43	22.01
				1999-2000	2.90	11.69	29.52
b. IFC Comprehensive Indexes							
1995-2000	8.20	21.25*	32.73*	1995-2000	20.00**	32.50**	43.17**
1995-1997	7.03	12.12	25.99	1995-1996	53.93**	58.50**	80.50**
1998-2000	6.46	17.26	24.32	1996-1997	10.02**	20.50*	31.12
				1997-1998	1.33	16.23	31.86*
				1998-1999	14.58**	6.03	22.41
				1999-2000	3.01	11.78	29.48

**, * Significant at the 1% and 5% level, respectively.

Table 3: Ljung-Box Q Statistics of the IFC Investable and Comprehensive Index Returns Series (Weekly and Daily) for the Czech Republic, Hungary, and Poland (Con't)

Period	Weekly			Period	Daily		
	Q(5)	Q(10)	Q(20)		Q(5)	Q(10)	Q(20)
Panel C: Poland							
a. IFC Investable Indexes							
1995-2000	2.86	5.57	14.08	1995-2000	40.86**	54.72**	67.78**
1995-1997	0.47	4.27	8.39	1995-1996	23.25**	32.97**	42.29**
1998-2000	2.52	4.81	13.39	1996-1997	35.72**	38.42**	45.70**
				1997-1998	10.38	14.62	26.38
				1998-1999	12.21*	20.89*	28.27
				1999-2000	3.88	8.53	16.59
b. IFC Comprehensive Indexes							
1995-2000	3.12	6.46	15.039	1995-2000	41.59**	55.45**	71.02**
1995-1997	1.45	8.81	19.17	1995-1996	31.14**	33.24**	41.43**
1998-2000	1.55	4.33	13.41	1996-1997	26.94**	31.45**	41.26**
				1997-1998	9.23	15.39	25.69
				1998-1999	10.25	18.11	34.76
				1999-2000	3.88	8.65	15.69

**, * Significant at the 1% and the 5% level, respectively.

Table 4: Unit Root Tests of IFC Investable and Comprehensive Indexes of Weekly Data

Index	ADF Test		PP Test	
	level series	first diff.	level series	first diff.
a. IFC Investable Indexes				
Czech Rep. (1)	-2.07	-9.70*	-2.43	-16.15*
Hungary (6)	-1.51	-6.23*	-1.45	-14.93*
Poland (1)	-2.03	-9.39*	-1.97	-13.89*
b. IFC Comprehensive Indexes				
Czech Rep. (2)	-1.91	-6.90*	-1.89	-13.69*
Hungary (6)	-1.61	-6.13*	-1.54	-14.44*
Poland (1)	-2.06	-9.10*	-1.99	-13.36*

*Significant at 5% level. Figures in parentheses indicate the number of lags used.

Table 5: Variance Ratios for Weekly Returns

		Number of weeks, k, in holding period				
		2	4	6	8	10
Czech Investable						
VR(k)		0.88	0.96	0.97	0.96	0.94
z(k)		-1.69	-0.33	-0.13	-0.17	-0.22
z *(k)		-0.99	-0.22	-0.05	-0.02	-0.03
Czech Comprehensive						
VR(k)		1.05	1.27	1.35	1.40	1.40
z (k)		0.81	2.09*	2.05*	1.93	1.71
z *(k)		0.55	1.46	0.51	0.26	0.20
Hungarian Investable						
VR(k)		0.97	1.16	1.27	1.23	1.16
z (k)		-0.42	1.23	1.59	1.14	0.77
z *(k)		-0.37	1.01	0.35	0.13	0.08
Hungarian Comprehensive						
VR(k)		1.01	1.23	1.35	1.33	1.28
z (k)		0.08	1.77	2.05*	1.62	1.20
z *(k)		0.06	1.46	0.47	0.20	0.13
Polish Investable						
VR(k)		1.05	1.14	1.18	1.16	1.16
z (k)		1.14	1.56	1.51	1.21	1.06
z *(k)		0.53	0.96	0.20	0.08	0.06
Polish Comprehensive						
VR(k)		1.07	1.20	1.26	1.25	1.25
z (k)		1.14	1.56	1.51	1.21	1.06
z *(k)		0.89	1.33	0.30	0.13	0.10

Under the null hypothesis $VR(k) = 1$, and the test statistics $z(k)$ and $z^*(k)$ are asymptotically distributed standard normal. The weekly data cover the period July 5, 1995, through September 27, 2000. (*) Variance ratios are different from 1 at the 5% level of significance.

Table 6: Johansen Cointegration Test Results for IFC Investable and Comprehensive Indexes

Null Hypothesis	Alternative Hypothesis	95% critical value	
Panel A: IFC Investable Indexes			
$\lambda_{\text{trace}}\text{test}$		$\lambda_{\text{trace}}\text{value}$	
$r = 0$	$r > 0$	22.32	31.52
$r \leq 1$	$r > 1$	11.07	17.95
$r \leq 2$	$r > 2$	5.14	8.18
Conclusion	$r = 0$		
$\lambda_{\text{max}}\text{test}$		$\lambda_{\text{max}}\text{value}$	
$r = 0$	$r = 1$	11.25	31.52
$r = 1$	$r = 2$	5.92	17.95
$r = 2$	$r = 2$	5.14	8.18
Conclusion	$r = 0$		
Panel B: IFC Comprehensive Indexes			
$\lambda_{\text{trace}}\text{test}$		$\lambda_{\text{trace}}\text{value}$	
$r = 0$	$r > 0$	26.17	31.52
$r \leq 1$	$r > 1$	12.23	17.95
$r \leq 2$	$r > 2$	5.75	8.18
Conclusion	$r = 0$		
$\lambda_{\text{max}}\text{test}$		$\lambda_{\text{max}}\text{value}$	
$r = 0$	$r = 1$	13.95	31.52
$r = 1$	$r = 2$	6.47	17.95
$r = 2$	$r = 2$	5.75	8.18
Conclusion	$r = 0$		

The λ_{trace} statistic tests the null of r cointegrating vectors against the alternative of more than r cointegrating relationships. The λ_{max} statistic tests the null hypothesis of r versus $r+1$ cointegrating vectors. The model includes a constant term and two lags as determined by the LR statistic. The 95% critical values are taken from Osterwald-Lenum (1992).

Table 7: Granger-Causality Tests of the Relationship between the Three Central European Stock Markets

a. IFC Investable Indexes

	F-statistic	Probability
Czech market does not cause Hungarian market	0.79	0.45
Hungarian market does not cause Czech market	2.81	0.06
Czech market does not cause Polish market	2.73	0.07
Polish market does not cause Czech market	1.23	0.29
Hungarian market does not cause Polish market	3.04*	0.05
Polish market does not cause Hungarian market	1.09	0.34

b. IFC Comprehensive Indexes

	F-statistic	Probability
Czech market does not cause Hungarian market	1.03	0.35
Hungarian market does not cause Czech market	1.71	0.18
Czech market does not cause Polish market	4.86*	0.01
Polish market does not cause Czech market	0.33	0.71
Hungarian market does not cause Polish market	1.68	0.18
Polish market does not cause Hungarian market	0.06	0.93

Indicates significant at the 5 percent level. The Granger-causality test is applied here to first differences of the index series pairwise. Since this test is highly sensitive to the lag orders of the right-hand-side variables, the Akaike criterion was used to determine the optimal lag length, which was two in each case.

Table 8: Forecasting Performance of Alternative Models

Week	RANDOM WALK		ARIMA		GARCH	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
Panel A: Czech Republic						
a. IFC Investable Indexes						
1	0.0371	0.0371	0.0288	0.0288	0.0303	0.0303
5	0.0424	0.0411	0.0434	0.0288	0.0434	0.0290
9	0.0436	0.0384	0.0390	0.0282	0.0390	0.0295
13	0.0410	0.0347	0.0382	0.0294	0.0382	0.0295
b. IFC Comprehensive Indexes						
1	0.0311	0.0311	0.0232	0.0232	0.0232	0.0232
5	0.0372	0.0363	0.0364	0.0232	0.0364	0.0233
9	0.0398	0.0350	0.0331	0.0239	0.0331	0.0240
13	0.0367	0.0315	0.0323	0.0248	0.0323	0.0249
Panel B: Hungary						
a. IFC Investable Indexes						
1	0.0661	0.0661	0.0576	0.0576	0.0576	0.0576
5	0.0660	0.0652	0.0298	0.0228	0.0298	0.0229
9	0.0882	0.0835	0.0276	0.0232	0.0276	0.0232
13	0.1379	0.1215	0.0280	0.0234	0.0280	0.0234
b. IFC Comprehensive Indexes						
1	0.0665	0.0665	0.0567	0.0567	0.0565	0.0565
5	0.0680	0.0672	0.0294	0.0226	0.0294	0.0226
9	0.0920	0.0870	0.0273	0.0231	0.0273	0.0231
13	0.1441	0.1268	0.0283	0.0242	0.0282	0.0242

The out-of-sample, one-week-ahead forecasts are dynamic forecasts, in that the sample base period is successfully expanded by the forecast values from the previous week(s). Results are shown only for weeks 1, 5, 9, and 13. The appropriate ARIMA and GARCH models are chosen according to the Akaike Information Criterion (AIC) and the Q Statistic for white-noise residuals.

Table 8: Forecasting Performance of Alternative Models (Con't)

Week	RANDOM WALK		ARIMA		GARCH	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
Panel C: Poland						
a. IFC Investable Indexes						
1	0.0047	0.0047	0.0016	0.0017	0.0019	0.0019
5	0.0254	0.0161	0.0246	0.0170	0.0246	0.0171
9	0.0580	0.0438	0.0234	0.0183	0.0234	0.0183
13	0.0867	0.0688	0.0273	0.0189	0.0273	0.0189
b. IFC Comprehensive Indexes						
1	0.0050	0.0050	0.0022	0.0022	0.0024	0.0024
5	0.0236	0.0158	0.0249	0.0176	0.0249	0.0176
9	0.0554	0.0420	0.0235	0.0186	0.0235	0.0186
13	0.0844	0.0668	0.0274	0.0191	0.0274	0.0191

The out-of-sample, one-week-ahead forecasts are dynamic forecasts, in that the sample base period is successfully expanded by the forecast values from the previous week(s). Results are shown only for weeks 1, 5, 9, and 13. The appropriate ARIMA and GARCH models are chosen according to the Akaike Information Criterion (AIC) and the Q Statistic for white-noise residuals.

Table 9: Significance Tests of the Forecasting Performance of Alternative Models

Panel A: Czech Republic						
	IFC Investable Indexes			IFC Comprehensive Indexes		
	RAN. WALK	ARIMA	GARCH	RAN. WALK	ARIMA	GARCH
RAN. WALK	—	0.011 (4.30*)	0.008 (2.64*)	—	0.006 (2.59*)	0.006 (2.60*)
ARIMA	-0.001 (-0.29)	—	—	-0.001 (-0.25)	—	—
GARCH	-0.002 (-1.80)	—	—	-0.001 (-0.25)	—	—
Panel B: Hungary						
	RAN. WALK	ARIMA	GARCH	RAN. WALK	ARIMA	GARCH
RAN. WALK	—	-0.012 (-6.08*)	-0.014 (-6.80*)	—	0.006 (2.60*)	-0.013 (-6.03*)
ARIMA	-0.002 (-1.79)	—	—	-0.001 (-0.25)	—	—
GARCH	-0.002 (-1.82)	—	—	-0.002 (-1.83)	—	—
Panel C: Poland						
	RAN. WALK	ARIMA	GARCH	RAN. WALK	ARIMA	GARCH
RAN. WALK	—	4.930 (6.77*)	2.411 (3.42*)	—	0.019 (2.81*)	0.846 (1.68)
ARIMA	-0.001 (-0.29)	—	—	-0.001 (-0.25)	—	—
GARCH	-0.001 (-0.29)	—	—	-0.001 (-0.25)	—	—

The dependent variables (forecasting errors) are shown in the column heading; the independent variables (forecasts) are shown in the row headings. The forecast variables are the log of stock prices.